

Sparse Propagation and Continuum Transfer Ledgers for MQM Thermal Response

Ben Mayes

MQM Research Development Draft, Version 0.1

Abstract—We construct the 298371-state two-excitation MQM bath matrix at 768 modes in sparse storage and formalize projection-defect bounds for its propagation. Exact-arithmetic short-step Krylov analysis separates finite propagation accuracy from the continuum comparison. A flat-band step-frequency embedding yields a rational second-order continuum-transfer allowance, but this coarse allowance remains insufficient for the original thermal target. Numerical controls validate sparse assembly and projection estimates without certifying floating-point propagation, the full covariance-weighted response, or hardware realization.

Index Terms—Quantum control, thermal response, sparse propagation, Krylov methods, continuum bounds.

I. SOURCE AND TYPED TARGET

The unchanged source has $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. For a finite bath, the fixed-launch second Gaussian response is

$$\Delta_2^M = \mathcal{R}_{T,2}(P_{p0} \otimes \Gamma_1(N)) - (\text{tr } N)\rho_0^M. \quad (1)$$

Here $\Gamma_1(N)$ is the one-photon occupation covariance and $\mathcal{R}_{T,2}$ propagates the joint system and bath before tracing the bath. Its numerical evaluation requires total-charge sector $Q = 2$, not closed evolution on the eight-dimensional retained matrix. The inherited sufficient second-order budget is $127/2^{25} - 2^{-128}$, exceeding that sufficient budget would not prove the full thermal target fails.

The certified finite-bath cutoffs from the parent specification are initial $K \leq 1$ and $K \leq 2$, with discarded weights below 2^{-23} and 2^{-36} at 768 modes. These are distinct from Gaussian perturbation orders and from propagation errors. They do not certify a uniform continuum cutoff.

II. SPARSE TWO-EXCITATION CONSTRUCTION

The basis consists of three native charge-two states, four blocks of M one-photon states, and a symmetric two-photon block with $M(M+1)/2$ states. Thus

$$D_2 = 3 + 4M + \binom{M+1}{2}. \quad (2)$$

The midpoint grid is $\delta_j = (2j+1-M)B/M$ and $c_j^2 = \kappa B/(\pi M)$. A pair $0 \leq j \leq k < M$ has zero-based index

$$p(j, k) = j(2M - j + 1)/2 + k - j. \quad (3)$$

The emission edge from $d1$ with photon j to the symmetric pair (j, k) has weight $c_k \sqrt{1 + \mathbf{1}_{j=k}}$. This bosonic factor is essential. Native clock edges and emission from the three native charge-two states are retained.

For even M , the undirected off-diagonal edge count is $M^2 + 6M + 2$. The diagonal has $4M + M(M+1)/2 - M/2$ nonzeros. At $M = 768 = 3 \cdot 2^8$ we obtain $D_2 = 298371$, 1486852 matrix nonzeros, and 19035712 CSR bytes using binary64 values and 32-bit indices. These are mathematical matrix resources, not physical qubit or interaction counts. No dense unitary is constructed.

At this grid, $\delta_j = (2j - 767)/48$, spacing is $1/24$, and $c_j^2 = 1/(6\pi)$. Free-bath relative phases align at 48π , and full phases at 96π . These arithmetic facts do not establish a coupled-latch recurrence or permanent retention.

III. PROJECTION DEFECT AND NUMERICAL LEDGER

Let H be the Hermitian finite-sector generator and v a normalized launch. An exact isometric Krylov basis V_k gives $T_k = V_k^\dagger H V_k$ and

$$\chi_k(t) = V_k e^{-iT_k t} e_1. \quad (4)$$

Define $R_k = H V_k - V_k T_k$. Duhamel's identity and unitarity yield

$$\|e^{-iHt} v - \chi_k(t)\| \leq \|v - V_k e_1\| + \int_0^t \|R_k e^{-iT_k s} e_1\| ds. \quad (5)$$

For exact Lanczos, $R_k = b_k v_{k+1} e_k^\dagger$, reducing the integral to $b_k \int_0^t |e_k^\dagger e^{-iT_k s} e_1| ds$. This defect methodology is prior art [1], not an MQM novelty claim.

For stored rounded matrices, use the actual full defect. If $R_k = b_k v_{k+1} e_k^\dagger + E$, add $t\|E\|$. Source-Hamiltonian mismatch, launch error, residual quadrature, and projected-exponential approximation must also be enclosed. A small numerical orthogonality residual is not such a certificate. Formula (5) also applies to any exactly interpreted stored V_k and Hermitian T_k ; isometry is needed for the ideal compression and polynomial bounds, not for Duhamel itself.

If a computed ket has vector error at most e , then its unnormalized density debit is at most $e + e^2/2$. Consequently finite-response propagation allowances add as

$$\delta_{\text{prop}} \leq \sum_j n_j (e_j + e_j^2/2) + (\text{tr } N)(e_0 + e_0^2/2). \quad (6)$$

Occupation-covariance rounding requires its own allowance. Numerical propagation and continuum transfer are separate ledger entries.

IV. IDEAL SHORT-STEP ALLOWANCE

A verified source-sector norm ceiling is 64. Exact Krylov compression matches powers through degree $k - 1$ and has norm at most 64. With step $\tau = 1/4$, $k = 64$, both exponential Taylor tails give the rational per-step bound

$$e_{\text{step}} \leq \frac{2 \cdot 16^{64}}{64!} \frac{1}{1 - 16/65}. \quad (7)$$

Sixty-four unitary steps reach $T = 16$ with total vector allowance below 2^{-32} , approximately 1.54949×10^{-10} . Using $\text{tr } N \leq 768/(2^{21} - 1)$ gives an ideal full finite-response propagation debit below 2^{-43} . This is a derived exact-arithmetic design. Restarted complex propagation and directed arithmetic certification have not been implemented. It does not report measured numerical accuracy or the actual thermal response.

V. CONDITIONAL CONTINUUM COMPARISON

Represent the continuum one-particle bath on $L^2([-B, B])$. Replace multiplication by δ with its midpoint step function δ_M . The flat coupling vector is constant, lies exactly in the bin-constant subspace, and retains norm $G^2 = \kappa B/\pi$. The step model preserves this subspace; its vacuum reduced dynamics equals the finite star model. On $Q \leq 2$,

$$\|H_\infty - H_{\text{step}}\| \leq d = 2B/M. \quad (8)$$

This is a frequency-operator comparison, not a photon-discard probability.

Let $n_* = 1/(2^{21} - 1)$ and $V_* = G^2 n_*$. The Bose occupation derivative is bounded by $\beta n_*(1 + n_*)$. Phase and occupation variations across a bin give

$$|C(s) - C_M(s)| \leq V_* \frac{B}{M} [\beta(1 + n_*) + s]. \quad (9)$$

In the interaction-picture double commutator, $\|S_s - S_s^M\| \leq 2sd$, each pair has trace-norm bound four, and the final unitary-channel difference is at most $2Td$. Thus the retained half-trace response difference obeys

$$\delta_{\text{cont},2} \leq 4 \int_{s>u} |C(s-u) - C_M(s-u)| dsdu \quad (10)$$

$$+ 8d \int_{s>u} (T + s + u) |C_M(s-u)| dsdu. \quad (11)$$

Since $|C_M| \leq V_*$ and the second geometric integral is T^3 , we obtain

$$\delta_{\text{cont},2} \leq V_* \frac{B}{M} \left[2\beta(1 + n_*)T^2 + \frac{50}{3}T^3 \right]. \quad (12)$$

The paired coefficient needs no sector above two. The comparison is conditional on the inherited reduced Gaussian reference; no divergent $\text{tr } N$ is taken separately in a continuum limit.

Positive degree-80 Taylor sums establish $e^{15} > 2^{21}$; $\pi > 3$ gives $G^2 < 128/3$. At 768 modes the exact rational majorant of (12) is

$$\frac{3444025262080}{118747142553627} \approx 0.0290030. \quad (13)$$

It is insufficient for the second-order budget. It is an upper allowance, not a lower bound on the actual response, and does not prove failure of the original specification.

VI. CONTROLS AND OUTSTANDING GATE

The frozen controls compare sparse assembly at $M = 16$ with an independent occupation-vector constructor. They test one hot-mode launch, with Krylov dimensions 16,32,64,128 over $T = 16$. Dense actual vector errors are approximately 1.719,1.672,0.6554,0.09618; all lie below their numerical full-defect estimates. Therefore those controls validate implementation, not target accuracy. A $k = 16$ pilot at $M = 768$ validates assembly and projection construction only. The full 768-column covariance-weighted response is not computed.

The short-step allowance was derived after exposure to these full-interval controls; the source, temperature, deadline, target, and interpretation rules were unchanged. Its algorithm design and implementation status are disclosed. The next gate is a sharper continuum-transfer enclosure together with certified restarted propagation. Physical promotion remains zero; no permanent hold, physical compiler, hardware performance, or comparative advantage is claimed.

REFERENCES

- [1] T. Jawecki, W. Auzinger, and O. Koch, "Computable upper error bounds for Krylov approximations to matrix exponentials and associated φ -functions," arXiv:1809.03369, 2019, doi:10.1007/s10543-019-00771-6.